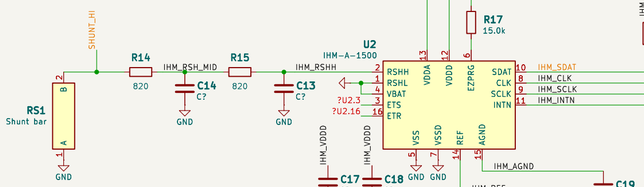
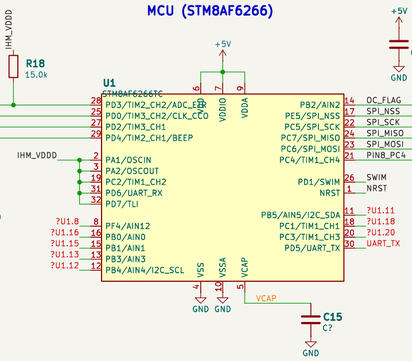
Current-sense input filter

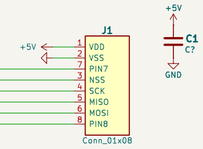


Current-sense ASIC (IHM-A-1500)

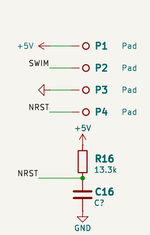
MCU (STM8AF6266)



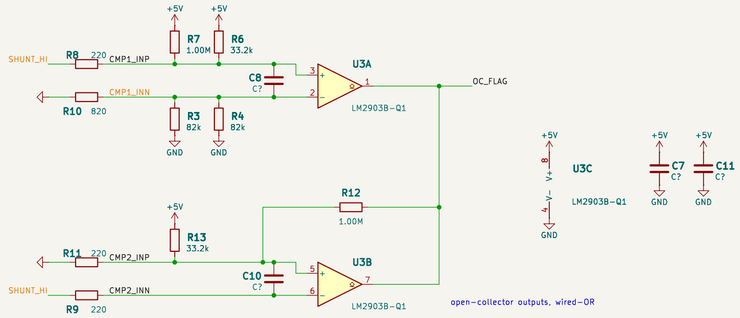
Harness connector



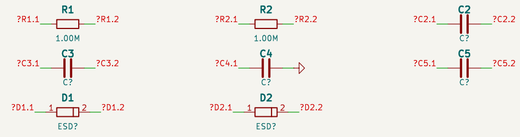
SWIM programming pads + reset



Bidirectional overcurrent detector (LM2903)



Connector-side parts (nets not measured yet)



Test pads



HOW TO READ: every pin carries a net label; identical label text = same net.
Green = measured / seen / datasheet. Orange = inferred. Red ?REF.PIN = destination unknown.
Passive pin 1 = top pad (vertical parts) or left pad (horizontal parts) in Traces-and-pins.jpeg.

SHUNT: board GND (and IHM RSHL, VBAT) is bonded to the bar. IHM RSHH sees the other end through a
2-pole RC filter (R14 820R + C14, then R15 820R + C13).
U3 = bidirectional overcurrent window. Each +IN = bar tap via 220R, offset toward +5V: +IN1 by R7 1M || R6 33.2k,
+IN2 by R13 33.2k (+33 mV) with R12 1M from OC_FLAG as hysteresis. Each -IN = bar tap via R10 820R / R9 220R. Wired-OR =
OC_FLAG (active low); connector PIN7 + MCU PB2, 2.7k pull-up R5. Trips at -34 mV across the shunt.
IHM is powered from MCU GPIOs U1.2/3/19/31/32 (switchable). ETS/ETR unused. Connector PIN8 = U1.21 (PC4).

Orange wire/label = inferred, red = unknown
Generated by gen_schematic.py — edit the CSVs, not this file

Sheet: /
File: shunt.kicad_sch

Title: Tesla Model S shunt board (RE)

Size: A2	Date: 2026-11-02	**Rev: auto**
KiCad E.D.A. 9.0.8		Id: 1/1